Austria Voter Transitions (R Markdown)

Introduction

This project uses Austria as a case study to estimate potential voter transitions between European Parliament political groups from the 2019 to the 2024 European Parliament election.

The analysis is based on electoral data from Austrian national political parties that contested the 2019 and 2024 European Parliament elections. Votes for each national party are classified according to the European Parliament political group to which the party belonged in the respective election year. The resulting political-group-level data are then used to estimate potential voter movements between European Parliament political groups and abstention, while allowing these estimated transitions to be interpreted in relation to the corresponding Austrian national parties.

Required Packages

The analysis uses four R packages. The tidyverse package is used to import, transform, organize, and manipulate the electoral data throughout the analysis. The scales package is used to format numerical values as percentages in the visualizations. The ggalluvial package is used to create the alluvial plot that visualizes selected estimated voter transitions. Finally, the sf package is used to import, process, and visualize the geographical data used in the regional electoral analysis.

```
library(tidyverse)
library(scales)
library(ggalluvial)
```

Import and Check Data

The electoral data used in this analysis were obtained from the official website of the Austrian Federal Ministry of the Interior (Bundesministerium für Inneres). The data contain the official results of the 2019 and 2024 European Parliament elections in Austria.

The data were organized at the NUTS 2 level, corresponding to Austria's nine federal states (Bundesländer): Burgenland, Carinthia, Lower Austria, Upper Austria, Salzburg, Styria, Tyrol, Vorarlberg, and Vienna. For each federal state, the datasets include the number of registered voters, votes received by the Austrian national parties, abstention, and invalid votes.

The original electoral results were organized and prepared in LibreOffice. Votes received by the Austrian national parties were then grouped according to the European Parliament political group to which each party belonged in the respective election year. The main national parties were classified as follows:

Austrian National Party	2019 European Political Group	2024 European Political Group
ÖVP	EPP	EPP
SPÖ	S&D	S&D
FPÖ	ID	PfE
The Greens	Greens/EFA	Greens/EFA
NEOS	Renew Europe	Renew Europe

Votes for parties outside these five main parties were combined into the Other (OTH) category. In 2019, OTH consists of KPÖ and EUROPA JETZT, while in 2024 it consists of KPÖ and DNA. The composition of this category therefore differs between the two elections.

In the prepared datasets, REG represents registered voters, ABS represents abstention, and INV represents invalid votes. The suffixes 19 and 24 indicate the 2019 and 2024 election years, respectively.

The prepared datasets were exported from LibreOffice as CSV files and subsequently imported into R for the voter-transition analysis.

```
austria2019 <- read_csv("austria_2019.csv")
austria2024 <- read_csv("austria_2024.csv")
```

The colnames() function is then used to inspect the variable names in both datasets before the data are transformed for the voter-transition analysis.

```
colnames(austria2024)
```

```
##  [1] "CODE"   "NUTS 2" "EPP24"  "SD24"   "PFE24"  "EFA24"  "RE
E24"  "OTH24"
##  [9] "REG24"  "ABS24"  "INV24"
```

```r
colnames(austria2019)
```

```
##  [1] "CODE"   "NUTS 2" "EPP19"  "SD19"   "ID19"   "EFA19"  "RE
E19"  "OTH19"
##  [9] "REG19"  "ABS19"  "INV19"
```

Prepare Inputs

The electoral data are transformed into the inputs required for the voter-transition estimation. First, the myN object is created using the number of registered voters in each federal state in 2019. This represents the size of the electorate in each regional unit and provides the population weights used by the VTR method.

```r
myN <- austria2019 %>% select(REG19)
```

The myX and myT objects represent the electoral distributions for 2019 and 2024, respectively, and are used as the two electoral-distribution inputs in the multirate() function. The number of votes in each category is divided by the number of registered voters in the corresponding federal state. This converts the original vote counts into proportions of the registered electorate, including abstention and invalid votes. Together with myN, these two objects are subsequently used by the VTR method to estimate voter transition rates between the two elections.

```r
myT <-
  austria2024 %>%
  mutate(
    EPP24 = EPP24 / REG24,
    SD24  = SD24 / REG24,
    PFE24 = PFE24 / REG24,
    EFA24 = EFA24 / REG24,
    REE24 = REE24 / REG24,
    OTH24 = OTH24 / REG24,
    ABS24 = ABS24 / REG24,
    INV24 = INV24 / REG24
  ) %>%
  select(EPP24, SD24, PFE24, EFA24, REE24, OTH24, ABS24, INV24)

myX <-
  austria2019 %>%
  mutate(
    EPP19 = EPP19 / REG19,
    SD19  = SD19 / REG19,
```

```
    ID19  = ID19 / REG19,
    EFA19 = EFA19 / REG19,
    REE19 = REE19 / REG19,
    OTH19 = OTH19 / REG19,
    ABS19 = ABS19 / REG19,
    INV19 = INV19 / REG19
  ) %>%
  select(EPP19, SD19, ID19, EFA19, REE19, OTH19, ABS19, INV19)
```

Data Checks

A series of consistency checks was conducted to verify that the federal-state codes matched correctly across the two datasets and that all transformed variables fell within the expected value range. Minimum and maximum values were also inspected to identify potential coding or transformation errors.

First, the federal-state codes were compared across the two datasets to verify that observations corresponded correctly between the two elections. The comparison was expected to return TRUE for all observations.

```
austria2019$CODE == austria2024$CODE
```

The total number of matching federal-state codes was then calculated as an additional check.

```
sum(austria2019$CODE == austria2024$CODE)
```

```
## [1] 9
```

All 9 federal-state codes matched across the two datasets.

Minimum and maximum values were inspected to ensure that all transformed variables remained within the valid proportional range from 0 to 1. Since myX and myT contain relative frequencies, each value represents the proportion of registered voters belonging to a particular electoral category and should therefore lie between 0 (0%) and 1 (100%).

```
myX %>%
  summarize_all(min) %>%
  print(width = Inf)

myX %>%
  summarize_all(max) %>%
  print(width = Inf)
```

```
myT %>%
  summarize_all(min) %>%
  print(width = Inf)

myT %>%
  summarize_all(max) %>%
  print(width = Inf)
```

The row sums of myX and myT were then inspected to verify that the electoral categories collectively accounted for the entire registered electorate in each federal state. Since myX and myT contain relative frequencies, the proportions across each row are expected to sum to approximately 1 (100%).

```
rowSums(myX)

## [1] 1 1 1 1 1 1 1 1 1

rowSums(myT)

## [1] 1 1 1 1 1 1 1 1 1
```

The row sums were approximately equal to 1 for all federal states, confirming that the electoral categories accounted for the full registered electorate.

As a final validation check, all values in the processed datasets were tested to confirm that they fell within the range from 0 to 1.

```
all(myX >= 0 & myX <= 1)

## [1] TRUE

all(myT >= 0 & myT <= 1)

## [1] TRUE
```

Both checks returned TRUE, confirming that all transformed values were within the expected proportional range.

Estimate Voter Transitions

The project requires the accompanying datasets and the external R script VTR.R to be located in the same working directory as the R Markdown file to ensure reproducibility.

The external script VTR.R was loaded to access the functions required for the estimation procedure.

```
source("VTR.R")
```

The multirate() function was then applied to myN, myX, and myT to estimate voter transition patterns between the 2019 and 2024 European Parliament elections in Austria. The value 0.01 was specified as the stopping threshold of the iterative estimation procedure, meaning that the algorithm continues until the remaining unallocated electorate falls to 1% or less.

```
z <- multirate(myN, myX, myT, 0.01)
```

```
## 1 ABS19 ABS24 0.9788681
## 2 EPP19 EPP24 0.6126849
## 3 SD19 SD24 0.7036286
## 4 ID19 PFE24 0.9899109
## 5 EFA19 EFA24 0.6677102
## 6 REE19 REE24 0.8212724
## 7 EPP19 PFE24 0.1361966
## 8 EPP19 ABS24 0.07199814
## 9 SD19 ABS24 0.145407
## 10 EFA19 OTH24 0.1442735
## 11 EPP19 SD24 0.05647896
## 12 OTH19 SD24 0.9763138
## 13 SD19 REE24 0.06936438
## 14 EFA19 ABS24 0.02118384
## 15 ABS19 PFE24 0.001544611
## 16 INV19 INV24 0.9274915
## 17 EPP19 EPP24 0.01532888
## 18 ABS19 PFE24 0.001550336
## 19 ABS19 OTH24 0.012614
## 20 EPP19 OTH24 0.02616202
## 21 EFA19 REE24 0.0157125
## 22 SD19 PFE24 0.008022916
## 23 SD19 OTH24 0.03127096
## 24 EFA19 OTH24 0.02309814
## 25 REE19 EFA24 0.01070997
## 26 EFA19 ABS24 0.01227667
## 27 ABS19 PFE24 0.0001842414
## 28 EPP19 REE24 0.005113711
## 29 EPP19 EPP24 0.009289402
## 30 EPP19 SD24 0.02492525
## 31 ABS19 PFE24 0.0002457387
## 32 EFA19 REE24 0.004822373
## 33 ABS19 PFE24 0.000216431
```

```
## 34 ABS19 PFE24 0.000189656
## 35 EPP19 ABS24 0.002450324
## 36 ABS19 PFE24 0.0001676355
## 37 ABS19 PFE24 0.0001503094
## 38 EPP19 PFE24 0.003933198
## 39 ABS19 PFE24 0.0001353909
## 40 ABS19 PFE24 0.0001230772
## 41 REE19 ABS24 0.01664572
## 42 REE19 PFE24 0.01396301
## 43 REE19 EFA24 0.00952004
## 44 ABS19 PFE24 0.0001151252
## 45 ABS19 PFE24 0.000105545
## 46 EFA19 PFE24 0.01377565
## 47 EFA19 EFA24 0.007546297
## 48 EFA19 REE24 0.003691766
## 49 EFA19 OTH24 0.005422073
## 50 ABS19 PFE24 0.0001014637
## 51 EPP19 INV24 0.001224094
## 52 ABS19 PFE24 9.532586e-05
## 53 SD19 PFE24 0.003625328
## 54 REE19 PFE24 0.009010322
## 55 EFA19 PFE24 0.006192289
## 56 EPP19 PFE24 0.002620114
## 57 EPP19 ABS24 0.001676366
## 58 EPP19 INV24 0.0009032211
## 59 EFA19 ABS24 0.005276833
## 60 ABS19 PFE24 8.201714e-05
## 61 ABS19 PFE24 7.596532e-05
## 62 EPP19 INV24 0.0007077062
## 63 ABS19 EPP24 9.823974e-05
## 64 ABS19 PFE24 6.809497e-05
## 65 EPP19 OTH24 0.0005209662
## 66 EPP19 EPP24 0.00211729
## 67 EFA19 EPP24 0.00533955
## 68 EFA19 OTH24 0.002779275
## 69 EFA19 PFE24 0.003478927
## 70 EFA19 REE24 0.001779816
## 71 EPP19 REE24 0.0007255485
## 72 ID19 ABS24 0.0005813515
## 73 ABS19 PFE24 7.117994e-05
## 74 ID19 ABS24 0.0004936536
## 75 ABS19 PFE24 6.631568e-05
## 76 ABS19 EPP24 8.383512e-05
## 77 ID19 ABS24 0.0004266591
## 78 ABS19 PFE24 5.903452e-05
## 79 SD19 SD24 0.001566361
```

```
## 80 EPP19 PFE24 0.001410623
## 81 EPP19 OTH24 0.0004024453
## 82 ABS19 EPP24 7.877952e-05
## 83 ID19 ABS24 0.0003849399
## 84 ID19 ABS24 0.0003390899
## 85 ABS19 EPP24 7.292212e-05
## 86 ABS19 PFE24 5.164141e-05
## 87 EFA19 OTH24 0.002083005
## 88 EPP19 REE24 0.0006008049
## 89 ID19 ABS24 0.0003142564
## 90 ABS19 EPP24 6.759999e-05
## 91 ABS19 PFE24 4.761256e-05
## 92 ID19 ABS24 0.0002809718
## 93 ABS19 EPP24 6.094869e-05
## 94 ABS19 PFE24 4.310017e-05
## 95 ID19 ABS24 0.0002531375
## 96 ABS19 PFE24 4.082466e-05
## 97 ABS19 EPP24 5.368682e-05
## 98 ID19 ABS24 0.0002295739
## 99 EPP19 ABS24 0.000859896
## 100 EPP19 PFE24 0.001036368
## 101 EPP19 INV24 0.0003628542
## 102 ABS19 EPP24 5.201016e-05
## 103 EFA19 REE24 0.001534713
## 104 EFA19 PFE24 0.002456152
## 105 ID19 ABS24 0.0002125565
## 106 EFA19 EFA24 0.002793368
## 107 ABS19 EPP24 5.195229e-05
## 108 EPP19 INV24 0.0003364443
## 109 ABS19 EPP24 4.92559e-05
## 110 ID19 ABS24 0.0002014287
## 111 EPP19 OTH24 0.0002843354
## 112 EPP19 EPP24 0.001164243
## 113 EPP19 OTH24 0.0002440646
## 114 ID19 ABS24 0.0001889373
## 115 ABS19 PFE24 3.404875e-05
## 116 ID19 ABS24 0.000173619
## 117 ABS19 PFE24 3.243762e-05
## 118 ABS19 EPP24 4.399426e-05
## 119 ID19 ABS24 0.0001603252
## 120 REE19 ABS24 0.01135417
## 121 REE19 PFE24 0.007524825
## 122 REE19 EPP24 0.00945103
## 123 REE19 EFA24 0.007762704
## 124 REE19 REE24 0.002921333
## 125 REE19 PFE24 0.003638492
```

```
## 126 REE19 OTH24 0.004091644
## 127 ABS19 EPP24 4.733246e-05
## 128 ABS19 EPP24 4.445816e-05
## 129 ID19 ABS24 0.0001639973
## 130 ABS19 EPP24 4.187128e-05
## 131 ID19 ABS24 0.0001512078
## 132 EPP19 ABS24 0.0006480262
## 133 SD19 ABS24 0.002727455
## 134 EFA19 ABS24 0.002543914
## 135 SD19 SD24 0.001521634
## 136 SD19 EPP24 0.003389878
## 137 EPP19 EPP24 0.00107175
## 138 EFA19 EPP24 0.002811883
## 139 REE19 ABS24 0.003666875
## 140 EPP19 ABS24 0.0004694139
## 141 SD19 EFA24 0.003238585
## 142 EPP19 INV24 0.0003472464
## 143 SD19 SD24 0.0007324121
## 144 ABS19 PFE24 3.560308e-05
## 145 ABS19 PFE24 3.357768e-05
## 146 ABS19 PFE24 3.174758e-05
## 147 ABS19 PFE24 3.008618e-05
## 148 ABS19 PFE24 2.857152e-05
## 149 EFA19 OTH24 0.001878886
## 150 EFA19 PFE24 0.002150027
## 151 EPP19 PFE24 0.0007492015
## 152 EPP19 EFA24 0.0009897205
## 153 EFA19 ABS24 0.001585869
## 154 EPP19 REE24 0.0004202524
## 155 ABS19 PFE24 2.688668e-05
## 156 SD19 INV24 0.0008095505
## 157 ABS19 PFE24 2.623066e-05
## 158 ID19 ABS24 0.0001438515
## 159 ABS19 EPP24 3.738354e-05
## 160 ABS19 PFE24 2.410125e-05
## 161 EPP19 PFE24 0.0005393692
## 162 EPP19 ABS24 0.000317534
## 163 REE19 PFE24 0.003458738
## 164 REE19 EFA24 0.005340392
## 165 REE19 REE24 0.002390548
## 166 ABS19 EPP24 3.823755e-05
## 167 ID19 ABS24 0.0001419058
## 168 SD19 REE24 0.001109296
## 169 EFA19 REE24 0.001250784
## 170 EPP19 REE24 0.0002956227
## 171 REE19 OTH24 0.002411183
```

```
## 172 REE19 PFE24 0.002231076
## 173 ABS19 EPP24 3.914309e-05
## 174 ID19 ABS24 0.0001461893
## 175 ID19 ABS24 0.0001337774
## 176 ABS19 EPP24 3.664204e-05
## 177 ID19 ABS24 0.0001231122
## 178 EPP19 INV24 0.0002009377
## 179 ID19 ABS24 0.000113913
## 180 ABS19 EPP24 3.466121e-05
## 181 ID19 ABS24 0.0001057016
## 182 ABS19 PFE24 1.963289e-05
## 183 ID19 ABS24 9.83842e-05
## 184 ABS19 PFE24 1.882203e-05
## 185 ABS19 EPP24 3.121138e-05
## 186 ID19 EFA24 0.0001821779
## 187 INV19 PFE24 0.0008166959
## 188 EFA19 PFE24 0.001601079
## 189 EFA19 EFA24 0.002399211
## 190 EPP19 EFA24 0.000699312
## 191 EPP19 SD24 0.0004803791
## 192 EPP19 EPP24 0.0006112874
## 193 EPP19 OTH24 0.000175277
## 194 ABS19 EPP24 3.228796e-05
## 195 ID19 ABS24 9.385558e-05
## 196 ID19 ABS24 8.757497e-05
## 197 ABS19 PFE24 1.760394e-05
## 198 ABS19 EPP24 2.973727e-05
## 199 ID19 ABS24 8.200324e-05
## 200 INV19 PFE24 0.0008289622
## 201 ID19 ABS24 7.696337e-05
## 202 EFA19 OTH24 0.001070108
## 203 EFA19 PFE24 0.001187794
## 204 SD19 SD24 0.0008798612
## 205 EPP19 SD24 0.0003527065
## 206 EPP19 ABS24 0.0002256329
## 207 EPP19 INV24 0.0001432484
## 208 INV19 PFE24 0.0008476791
## 209 ABS19 PFE24 1.712908e-05
## 210 INV19 PFE24 0.0008050317
## 211 EPP19 INV24 0.0001271327
## 212 REE19 SD24 0.00374455
## 213 EFA19 PFE24 0.001107593
## 214 EPP19 PFE24 0.0002587545
## 215 REE19 EPP24 0.003632916
## 216 REE19 REE24 0.001598908
## 217 SD19 EPP24 0.002524442
```

```
## 218 EPP19 EPP24 0.0004185054
## 219 SD19 REE24 0.0009595312
## 220 EPP19 REE24 0.0001746455
## 221 ID19 ABS24 8.852889e-05
## 222 EFA19 REE24 0.0008356928
## 223 EFA19 EPP24 0.001559522
## 224 EFA19 OTH24 0.0009278654
## 225 ID19 ABS24 8.730036e-05
## 226 REE19 ABS24 0.002085524
## 227 EPP19 ABS24 0.0002049655
## 228 EFA19 ABS24 0.0009282793
## 229 SD19 ABS24 0.0009235005
## 230 SD19 INV24 0.0006484129
```

The output of the multirate() function is stored in the object z, which contains the estimated voter-transition results, including the bb and Bb components. The z$bb component contains transition estimates for the individual regional units, while z$Bb contains the aggregate voter-transition matrix with the estimated transition rates across Austria. Since this analysis focuses on aggregate voter-transition patterns, z$Bb is used in the subsequent analysis.

The complete z object is then saved as an RDS file and subsequently reloaded, allowing the estimated results to be stored and reused without rerunning the multirate() procedure.

```
write_rds(z, "z.rds")
z <- read_rds("z.rds")
```

Export Aggregate Transition Matrix

The aggregate voter-transition matrix contained in z$Bb was exported as a CSV file to provide a separate record of the estimated transition rates. Since the 2019 electoral categories are stored as row names in the matrix, the matrix was first converted into a data frame and the row names were converted into a regular column before export.

```
z$Bb %>%
  as.data.frame() %>%
  rownames_to_column() %>%
  write_csv2(file = "austria19-austria24.csv")
```

Prepare Transition Results

The aggregate voter-transition matrix z$Bb was used as the basis for preparing the transition results for further analysis. The matrix was first converted into a data frame using the as.data.frame() function. The rownames_to_column() function was then used to convert the row names, which represent the 2019 electoral categories, into a regular column named group_2019.

The pivot_longer() function was subsequently used to transform the data from wide to long format. The resulting long-format data frame contains three columns: group_2019, which identifies the 2019 electoral category; group_2024, which identifies the 2024 destination category; and transition_rate, which contains the corresponding estimated voter-transition rate. Finally, the arrange() function was used to order the estimated voter transitions from the highest to the lowest transition rate.

```
transitions_long <-
  z$Bb %>%
  as.data.frame() %>%
  rownames_to_column("group_2019") %>%
  pivot_longer(
    cols = -group_2019,
    names_to = "group_2024",
    values_to = "transition_rate"
  ) %>%
  arrange(desc(transition_rate))

transitions_long
```

```
## # A tibble: 64 × 3
##    group_2019 group_2024 transition_rate
##    <chr>      <chr>                <dbl>
##  1 ID19       PFE24                0.990
##  2 ABS19      ABS24                0.979
##  3 OTH19      SD24                 0.976
##  4 INV19      INV24                0.927
##  5 REE19      REE24                0.828
##  6 SD19       SD24                 0.708
##  7 EFA19      EFA24                0.680
##  8 EPP19      EPP24                0.643
##  9 EFA19      OTH24                0.182
## 10 SD19       ABS24                0.149
## # ℹ 54 more rows
```

Filter Major Voter Transitions

To focus the subsequent analysis on the more substantial estimated voter movements, the filter() function was applied to the transitions_long data frame. Only estimated voter transitions with a transition_rate greater than or equal to 0.05 (5%) were retained. The resulting observations were stored in the major_transitions object.

```
major_transitions <-
  transitions_long %>%
  filter(transition_rate >= 0.05)

major_transitions
```

```
## # A tibble: 14 × 3
##    group_2019 group_2024 transition_rate
##    <chr>      <chr>                <dbl>
##  1 ID19       PFE24               0.990
##  2 ABS19      ABS24               0.979
##  3 OTH19      SD24                0.976
##  4 INV19      INV24               0.927
##  5 REE19      REE24               0.828
##  6 SD19       SD24                0.708
##  7 EFA19      EFA24               0.680
##  8 EPP19      EPP24               0.643
##  9 EFA19      OTH24               0.182
## 10 SD19       ABS24               0.149
## 11 EPP19      PFE24               0.147
## 12 EPP19      SD24                0.0822
## 13 EPP19      ABS24               0.0789
## 14 SD19       REE24               0.0714
```

Voter Retention

To examine estimated voter retention, five transitions were selected from the major_transitions data frame using the filter() function. These correspond to EPP → EPP, S&D → S&D, Greens/EFA → Greens/EFA, Renew → Renew, and ID → PfE. The first four represent retention within the same European Parliament political group, while ID → PfE reflects the change in European Parliament political-group affiliation of FPÖ between the two elections.

The mutate() function was then used to create two additional descriptive columns, with case_when() assigning labels according to the values of group_2019. The european_political_groups column provides readable labels for the selected European political-group transitions, while the national_party column identifies the corresponding Austrian national party. In this way, EPP is associated with ÖVP, S&D with SPÖ, Greens/EFA with the Greens, Renew with NEOS, and the ID → PfE transition with FPÖ. The resulting observations were stored in the retention object.

```
retention <-
  major_transitions %>%
  filter(
    (group_2019 == "ID19"  & group_2024 == "PFE24") |
      (group_2019 == "REE19" & group_2024 == "REE24") |
      (group_2019 == "SD19"  & group_2024 == "SD24") |
      (group_2019 == "EFA19" & group_2024 == "EFA24") |
      (group_2019 == "EPP19" & group_2024 == "EPP24")
  ) %>%
  mutate(
    european_political_groups = case_when(
      group_2019 == "ID19"  ~ "ID → PfE",
      group_2019 == "REE19" ~ "Renew",
      group_2019 == "SD19"  ~ "S&D",
      group_2019 == "EFA19" ~ "Greens/EFA",
      group_2019 == "EPP19" ~ "EPP"
    ),
    national_party = case_when(
      group_2019 == "ID19"  ~ "FPÖ",
      group_2019 == "REE19" ~ "NEOS",
      group_2019 == "SD19"  ~ "SPÖ",
      group_2019 == "EFA19" ~ "Greens",
      group_2019 == "EPP19" ~ "ÖVP"
    )
  )

retention
```

```
## # A tibble: 5 × 5
##   group_2019 group_2024 transition_rate european_political_gro
ups national_party
##   <chr>      <chr>                <dbl> <chr>
<chr>
## 1 ID19       PFE24                0.990 ID → PfE
FPÖ
## 2 REE19      REE24                0.828 Renew
```

```
NEOS
## 3 SD19       SD24                  0.708 S&D
SPÖ
## 4 EFA19      EFA24                 0.680 Greens/EFA
Greens
## 5 EPP19      EPP24                 0.643 EPP
ÖVP
```

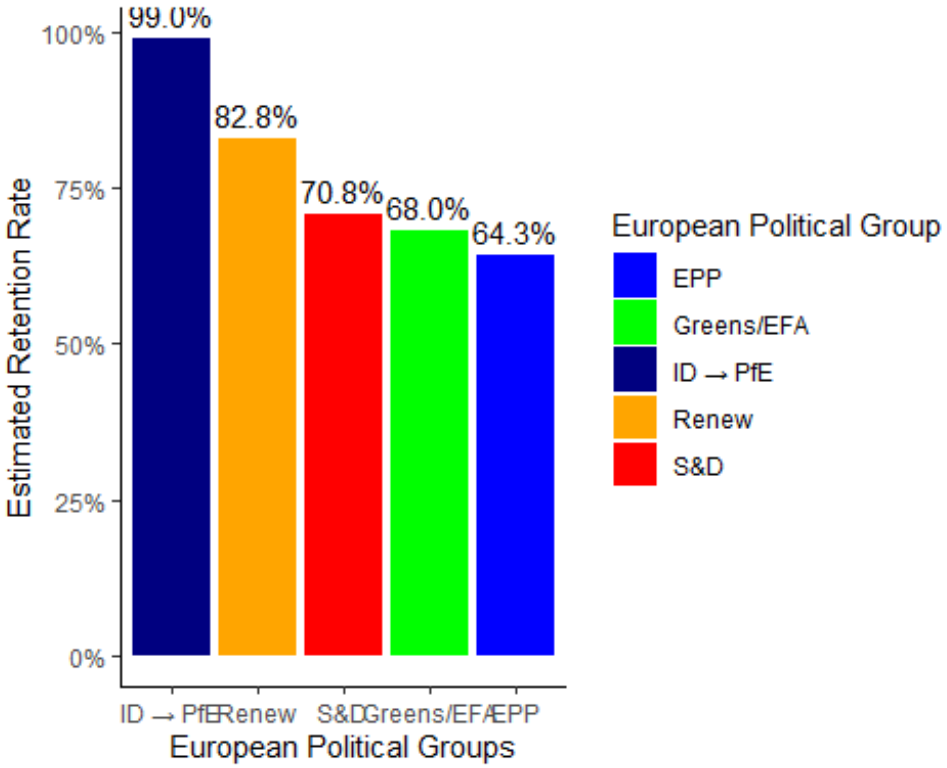
Voter Retention Visualization

The estimated retention rates stored in the retention object were visualized using ggplot(). The European Parliament political groups were placed on the x-axis, while the corresponding estimated retention rates were placed on the y-axis. The reorder() function was used to arrange the political groups from the highest to the lowest estimated retention rate.

The geom_col() function was used to represent the retention rates as vertical bars, while geom_text() was used to display the estimated percentage above each bar. The scale_fill_manual() function assigned a distinct color to each European Parliament political group, and scale_y_continuous() formatted the values on the y-axis as percentages. Finally, labs() was used to define the axis and legend labels, and theme_classic() was applied to provide a clean visual presentation.

```
retention %>%
  ggplot(aes(
    x = reorder(european_political_groups, -transition_rate),
    y = transition_rate,
    fill = european_political_groups
  )) +
  geom_col() +
  geom_text(
    aes(label = scales::percent(transition_rate, accuracy = 0.1))
,
    vjust = -0.5
  ) +
  scale_fill_manual(
    values = c(
      "ID → PfE" = "navy",
      "Renew" = "orange",
      "S&D" = "red",
      "Greens/EFA" = "green",
      "EPP" = "blue"
    )
```

```
  ) +
scale_y_continuous(labels = scales::label_percent()) +
labs(
  x = "European Political Groups",
  y = "Estimated Retention Rate",
  fill = "European Political Group"
) +
theme_classic()
```



Selected Voter Transitions

To examine selected estimated voter movements, four transitions were selected from the major_transitions data frame using the filter() function. These correspond to EPP → PfE, EPP → S&D, EPP → Abstention, and S&D → Abstention. These transitions were selected to examine notable estimated movements from EPP and S&D towards other European Parliament political groups or abstention between the two elections.

The mutate() function was then used to create two additional descriptive columns, with case_when() assigning labels according to the combinations of group_2019 and group_2024. The european_group_transition column provides readable labels for the selected transitions, while the national_party

column provides the corresponding interpretation in terms of Austrian national parties. In this way, EPP → PfE is associated with ÖVP → FPÖ, EPP → S&D with ÖVP → SPÖ, EPP → Abstention with ÖVP → Abstention, and S&D → Abstention with SPÖ → Abstention. The resulting observations were stored in the selected_transitions object.

```
selected_transitions <-
  major_transitions %>%
  filter(
    (group_2019 == "EPP19" & group_2024 == "PFE24") |
      (group_2019 == "EPP19" & group_2024 == "SD24") |
      (group_2019 == "EPP19" & group_2024 == "ABS24") |
      (group_2019 == "SD19"  & group_2024 == "ABS24")
  ) %>%
  mutate(
    european_group_transition = case_when(
      group_2019 == "EPP19" & group_2024 == "PFE24" ~ "EPP → PfE"
,
      group_2019 == "EPP19" & group_2024 == "SD24"  ~ "EPP → S&D"
,
      group_2019 == "EPP19" & group_2024 == "ABS24" ~ "EPP → Abst
ention",
      group_2019 == "SD19"  & group_2024 == "ABS24" ~ "S&D → Abst
ention"
    ),
    national_party = case_when(
      group_2019 == "EPP19" & group_2024 == "PFE24" ~ "ÖVP → FPÖ"
,
      group_2019 == "EPP19" & group_2024 == "SD24"  ~ "ÖVP → SPÖ"
,
      group_2019 == "EPP19" & group_2024 == "ABS24" ~ "ÖVP → Abst
ention",
      group_2019 == "SD19"  & group_2024 == "ABS24" ~ "SPÖ → Abst
ention"
    )
  )

selected_transitions
```

```
## # A tibble: 4 × 5
##   group_2019 group_2024 transition_rate european_group_transit
ion national_party
##   <chr>      <chr>                <dbl> <chr>
<chr>
## 1 SD19       ABS24                0.149  S&D → Abstention
```

```
SPÖ → Abstent…
## 2 EPP19      PFE24                0.147  EPP → PfE
ÖVP → FPÖ
## 3 EPP19      SD24                 0.0822 EPP → S&D
ÖVP → SPÖ
## 4 EPP19      ABS24                0.0789 EPP → Abstention
ÖVP → Abstent…
```

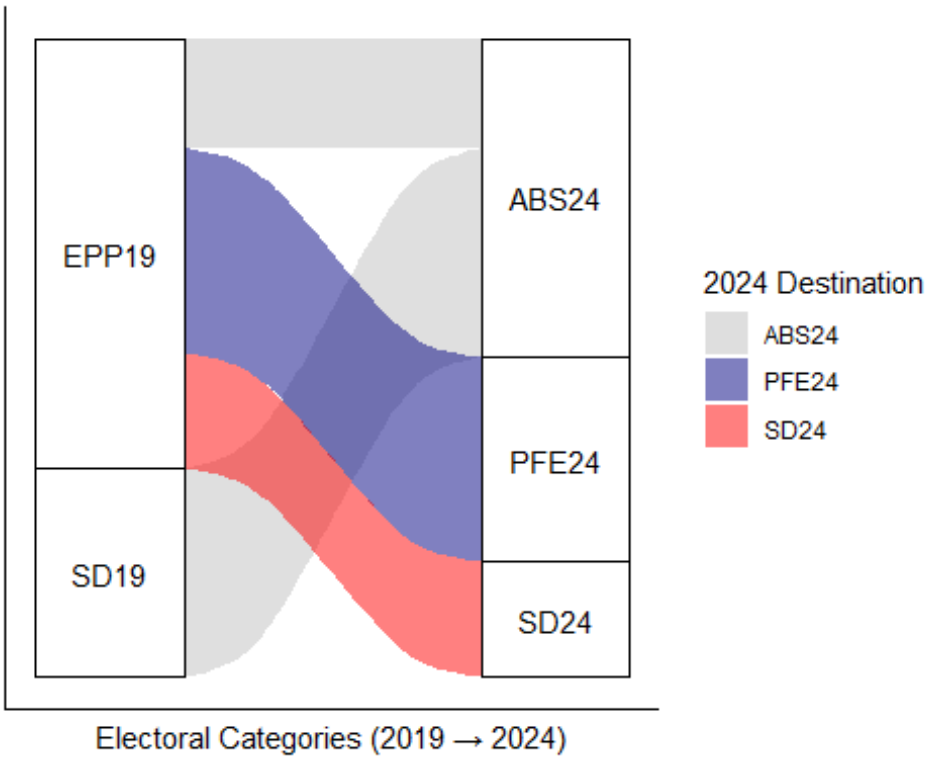
Voter Transition Visualization

The selected estimated voter transitions stored in the selected_transitions object were visualized using ggplot() and an alluvial plot. The 2019 electoral categories were placed on the left side of the plot and the 2024 destination categories on the right side, while transition_rate determines the size of each estimated voter flow.

The geom_alluvium() function was used to represent the estimated voter movements between the two elections, with the flows colored according to their 2024 destination category. The geom_stratum() function was used to display the electoral categories on both sides of the plot, while geom_text() was used to label these categories. The scale_fill_manual() function assigned distinct colors to PfE, S&D, and Abstention. The labs() function was used to define the axis and legend labels, while the y-axis label was removed. Finally, theme_classic() and additional theme() settings were used to remove unnecessary axis text and ticks and provide a cleaner visual presentation.

```r
selected_transitions %>%
  ggplot(aes(
    axis1 = group_2019,
    axis2 = group_2024,
    y = transition_rate
  )) +
  geom_alluvium(aes(fill = group_2024)) +
  geom_stratum() +
  geom_text(
    stat = "stratum",
    aes(label = after_stat(stratum))
  ) +
  scale_fill_manual(
    values = c(
      "PFE24" = "navy",
      "SD24" = "red",
      "ABS24" = "grey"
```

```
  )
) +
labs(
  x = "Electoral Categories (2019 → 2024)",
  y = NULL,
  fill = "2024 Destination"
) +
theme_classic() +
theme(
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank(),
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank()
)
```



Regional Electoral Analysis

Geographical boundaries for Austria's nine NUTS-2 regions were obtained from Eurostat's GISCO geographical database. The NUTS 2024 Level 2 regional polygons were downloaded in GeoJSON format at a 1:10 million scale and using the EPSG:4326 coordinate reference system. The file was saved in the project directory as austria_map.geojson and imported into R

using the sf package. These geographical boundaries were used to visualize regional changes in electoral support between the 2019 and 2024 European Parliament elections.

```
library(sf)

austria_map <- st_read(
  "austria_map.geojson",
  quiet = TRUE
)
```

The geographical dataset contains NUTS-2 regions from multiple European countries. Therefore, the data were filtered using the country code to retain only the Austrian NUTS-2 regions.

```
austria_map <-
  austria_map %>%
  filter(CNTR_CODE == "AT")
```

The resulting spatial dataset contains the nine NUTS-2 regions of Austria. The regional codes and names were inspected before combining the geographical boundaries with the electoral data.

```
austria_map %>%
  select(NUTS_ID, NAME_LATN) %>%
  st_drop_geometry()
```

```
##   NUTS_ID        NAME_LATN
## 1    AT11       Burgenland
## 2    AT12 Niederösterreich
## 3    AT13             Wien
## 4    AT21          Kärnten
## 5    AT22       Steiermark
## 6    AT31   Oberösterreich
## 7    AT32         Salzburg
## 8    AT33            Tirol
## 9    AT34       Vorarlberg
```

Regional Change in EPP/ÖVP Electoral Support

To examine regional differences in electoral support, the vote share of the EPP/ÖVP was calculated for each Austrian NUTS-2 region in both elections. Valid votes were calculated by subtracting abstentions and invalid votes from the number of registered voters. The change in electoral support was then expressed in percentage points between 2019 and 2024.

```r
epp_regional <-
  austria2019 %>%
  select(CODE, `NUTS 2`, EPP19, REG19, ABS19, INV19) %>%
  left_join(
    austria2024 %>%
      select(CODE, EPP24, REG24, ABS24, INV24),
    by = "CODE"
  ) %>%
  mutate(
    valid19 = REG19 - ABS19 - INV19,
    valid24 = REG24 - ABS24 - INV24,
    EPP19_share = EPP19 / valid19,
    EPP24_share = EPP24 / valid24,
    EPP_change = (EPP24_share - EPP19_share) * 100
  )

epp_regional
```

```
## # A tibble: 9 × 15
##   CODE  `NUTS 2`    EPP19  REG19  ABS19 INV19  EPP24  REG24  A
BS24 INV24 valid19
##   <chr> <chr>       <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>  <
dbl> <dbl>   <dbl>
## 1 AT11  Burgenland  53809 2.35e5  79245  3406  37654 2.35e5  9
7454  3131  151958
## 2 AT12  Niederöst… 342105 1.30e6 425019 16744 234155 1.30e6 47
1521 18529  853764
## 3 AT13  Wien       144073 1.16e6 477972  6056  84528 1.14e6 53
7575  6179  673653
## 4 AT21  Kärnten     64841 4.39e5 210486  4170  40839 4.34e5 21
3964  4197  224383
## 5 AT22  Steiermark 193772 9.66e5 418252  5656 132794 9.52e5 42
0695  6795  542211
## 6 AT31  Oberöster… 237585 1.11e6 420170 12070 159927 1.10e6 44
5827 14481  676208
## 7 AT32  Salzburg    97376 3.96e5 167115  3105  61170 3.92e5 18
1121  3302  226007
## 8 AT33  Tirol      122240 5.44e5 254145  2622  76348 5.41e5 28
2199  2901  286765
## 9 AT34  Vorarlberg  50155 2.75e5 129111  1069  36657 2.77e5 13
7393  1006  144815
## # ℹ 4 more variables: valid24 <dbl>, EPP19_share <dbl>, EPP24_
share <dbl>,
## #   EPP_change <dbl>
```

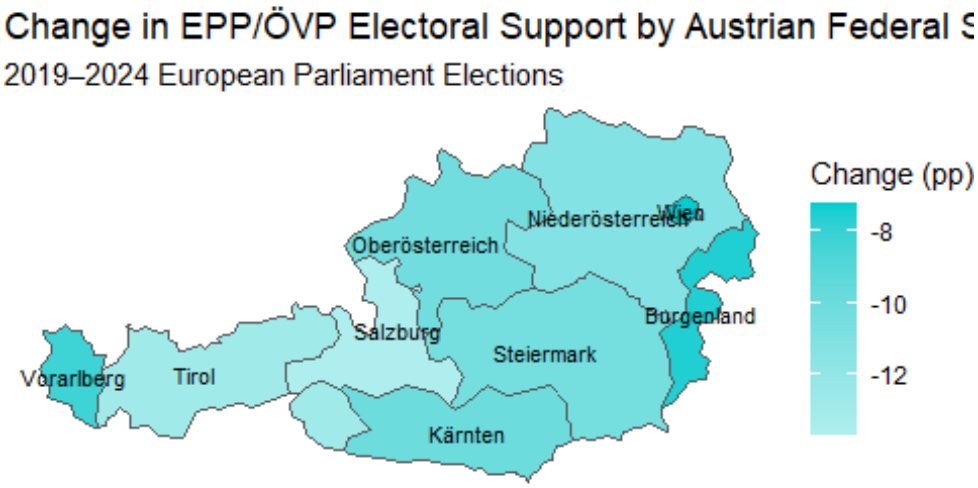
Regional Distribution of Changes in EPP/ÖVP Support

To visualize the regional differences in EPP/ÖVP electoral support, the electoral data were combined with Austria's NUTS-2 geographical boundaries. The NUTS_ID variable in the geographical data corresponds to the CODE variable in the electoral data, allowing each region's electoral change to be matched with its geographical area.

```
epp_map <-
  austria_map %>%
  left_join(
    epp_regional %>%
      select(CODE, EPP_change),
    by = c("NUTS_ID" = "CODE")
  )

epp_map_projected <- st_transform(epp_map, 3035)

ggplot(epp_map_projected) +
  geom_sf(aes(fill = EPP_change)) +
  geom_sf_text(
    aes(label = NAME_LATN),
    size = 3
  ) +
  scale_fill_gradient(
    low = "paleturquoise",
    high = "darkturquoise",
    name = "Change (pp)"
  ) +
  labs(
    title = "Change in EPP/ÖVP Electoral Support by Austrian Fede
ral State",
    subtitle = "2019–2024 European Parliament Elections",
    fill = "Change (pp)"
  ) +
  theme_void()
```



The map displays the change in the EPP/ÖVP share of valid votes between the 2019 and 2024 European Parliament elections, measured in percentage points rather than estimated voter transitions. Lighter shades indicate larger declines in EPP/ÖVP electoral support, while darker shades indicate smaller declines. Therefore, the map provides a descriptive regional comparison of electoral support and complements the Austria-wide voter transition estimates produced by the VTR analysis.

ÖVP Regional Vote Share by Austrian Federal State

The following table presents the ÖVP's vote share in each Austrian federal state in the 2019 and 2024 European Parliament elections, together with the change in percentage points.

```
epp_table <-
  epp_regional %>%
  select(`NUTS 2`, EPP19_share, EPP24_share, EPP_change) %>%
  arrange(EPP_change)

epp_table
```

```
## # A tibble: 9 × 4
##   `NUTS 2`          EPP19_share EPP24_share EPP_change
##   <chr>                   <dbl>       <dbl>      <dbl>
## 1 Salzburg                0.431       0.294     -13.7
## 2 Tirol                   0.426       0.298     -12.8
## 3 Niederösterreich        0.401       0.289     -11.1
## 4 Steiermark              0.357       0.253     -10.4
## 5 Oberösterreich          0.351       0.249     -10.2
## 6 Kärnten                 0.289       0.189      -9.95
## 7 Vorarlberg              0.346       0.265      -8.17
## 8 Burgenland              0.354       0.279      -7.50
## 9 Wien                    0.214       0.142      -7.20
```

```
write_csv(epp_table, "ovp_regional_results.csv")
```

The table is ordered from the largest to the smallest decline in ÖVP vote share. The results are based directly on the official election results and therefore represent actual changes in vote share rather than estimated voter transitions.

Regional Change in FPÖ Support

To examine the geographical variation in FPÖ support, the change in the party's vote share between the 2019 and 2024 European Parliament elections was calculated across Austria's nine federal states. The comparison is based on the official election results rather than the VTR estimates. For each election, the number of votes received by the FPÖ was divided by the total number of valid votes in the respective federal state. The change in support was then calculated by subtracting the FPÖ's 2019 vote share from its 2024 vote share and expressing the difference in percentage points.

```
FPO_regional <-
  austria2019 %>%
  select(CODE, `NUTS 2`, ID19, REG19, ABS19, INV19) %>%
  left_join(
    austria2024 %>%
      select(CODE, PFE24, REG24, ABS24, INV24),
    by = "CODE"
  ) %>%
  mutate(
    valid19 = REG19 - ABS19 - INV19,
    valid24 = REG24 - ABS24 - INV24,
    ID19_share = ID19 / valid19,
    PFE24_share = PFE24 / valid24,
    FPO_change = (PFE24_share - ID19_share) * 100
```

```
  )

FPO_regional
```

```
## # A tibble: 9 × 15
##   CODE  `NUTS 2`     ID19  REG19  ABS19 INV19  PFE24  REG24  A
BS24 INV24 valid19
##   <chr> <chr>       <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>  <
dbl> <dbl>   <dbl>
## 1 AT11  Burgenland  26649 2.35e5  79245  3406  33799 2.35e5  9
7454  3131  151958
## 2 AT12  Niederöst… 152023 1.30e6 425019 16744 213678 1.30e6 47
1521 18529  853764
## 3 AT13  Wien        97010 1.16e6 477972  6056 107800 1.14e6 53
7575  6179  673653
## 4 AT21  Kärnten     48400 4.39e5 210486  4170  71469 4.34e5 21
3964  4197  224383
## 5 AT22  Steiermark 107040 9.66e5 418252  5656 145557 9.52e5 42
0695  6795  542211
## 6 AT31  Oberöster… 122263 1.11e6 420170 12070 178529 1.10e6 44
5827 14481  676208
## 7 AT32  Salzburg    32845 3.96e5 167115  3105  50429 3.92e5 18
1121  3302  226007
## 8 AT33  Tirol       43487 5.44e5 254145  2622  61228 5.41e5 28
2199  2901  286765
## 9 AT34  Vorarlberg  20397 2.75e5 129111  1069  31264 2.77e5 13
7393  1006  144815
## # ℹ 4 more variables: valid24 <dbl>, ID19_share <dbl>, PFE24_s
hare <dbl>,
## #   FPO_change <dbl>
```

The variable FPO_change measures the change in the FPÖ's vote share between the two elections in percentage points. Positive values indicate an increase in electoral support.

Geographical Distribution of FPÖ Vote-Share Change

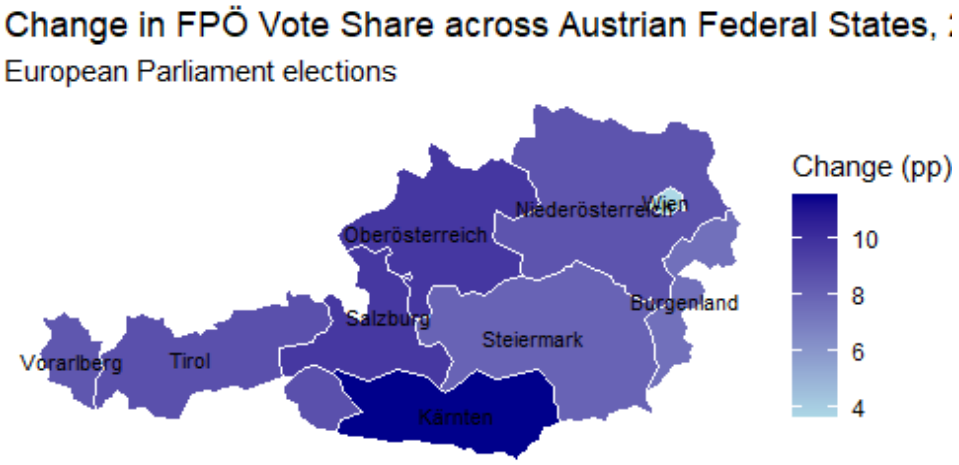
The regional variation in the FPÖ's electoral gains is visualised using Austria's NUTS 2 boundaries. Darker shades of blue indicate larger increases in the FPÖ's vote share between 2019 and 2024.

```
FPO_map <- austria_map %>%
  left_join(
    FPO_regional %>%
      select(CODE, FPO_change),
    by = c("NUTS_ID" = "CODE")
```

```r
  )

FPO_map_projected <- st_transform(FPO_map, 3035)

ggplot(FPO_map_projected) +
  geom_sf(
    aes(fill = FPO_change),
    color = "white",
    linewidth = 0.5
  ) +
  geom_sf_text(
    aes(label = NAME_LATN),
    size = 3
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue",
    name = "Change (pp)"
  ) +
  labs(
    title = "Change in FPÖ Vote Share across Austrian Federal Sta
tes, 2019–2024",
    subtitle = "European Parliament elections"
  ) +
  theme_void()
```



FPÖ Regional Vote Share by Austrian Federal State

The following table presents the FPÖ's vote share in each Austrian federal state in the 2019 and 2024 European Parliament elections, together with the change in percentage points.

```
FPO_table <-
  FPO_regional %>%
  select(`NUTS 2`, ID19_share, PFE24_share, FPO_change) %>%
  arrange(desc(FPO_change))

FPO_table
```

```
## # A tibble: 9 × 4
##   `NUTS 2`          ID19_share PFE24_share FPO_change
##   <chr>                  <dbl>       <dbl>      <dbl>
## 1 Kärnten                0.216       0.332      11.6
## 2 Oberösterreich         0.181       0.278       9.74
## 3 Salzburg               0.145       0.243       9.72
## 4 Tirol                  0.152       0.239       8.76
## 5 Niederösterreich       0.178       0.264       8.60
## 6 Vorarlberg             0.141       0.226       8.49
## 7 Steiermark             0.197       0.278       8.03
## 8 Burgenland             0.175       0.251       7.52
## 9 Wien                   0.144       0.181       3.69
```

```
write_csv(FPO_table, "fpo_regional_results.csv")
```

The table is ordered from the largest to the smallest increase in FPÖ vote share. The results are based directly on the official election results and therefore represent actual changes in vote share rather than estimated voter transitions.